Public Transit and Socio-Spatial Exclusion in Toronto, Canada

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Introduction

Social exclusion can be understood as a complex and dynamic set of processes and relationships that prevent individuals and groups from accessing resources, participating in society, and asserting their rights (Beall and Prion, 2004). Social exclusion is present at varying degrees in almost all cities around the world, and it often also spatially or geographically distributed. Socio-spatial exclusion can lead to various negative consequences for a city, including marginalization of certain groups, vicious cycles of unemployment, poverty, crime, and poor physical and mental health. It is important for municipal governments, policy makers, and city planners to address issues of socio-spatial exclusion in order to maintain a prosperous and sustainable city. In this paper, I will examine issues of socio-spatial exclusion in the city of Toronto, Canada. Using maps to illustrate the geographic distribution of different socio-economic factors including wealth and health across the city, I seek to provide insight into how social issues are spatially distributed in Toronto, as well as how access to public transit affects these inequalities.

Toronto Context

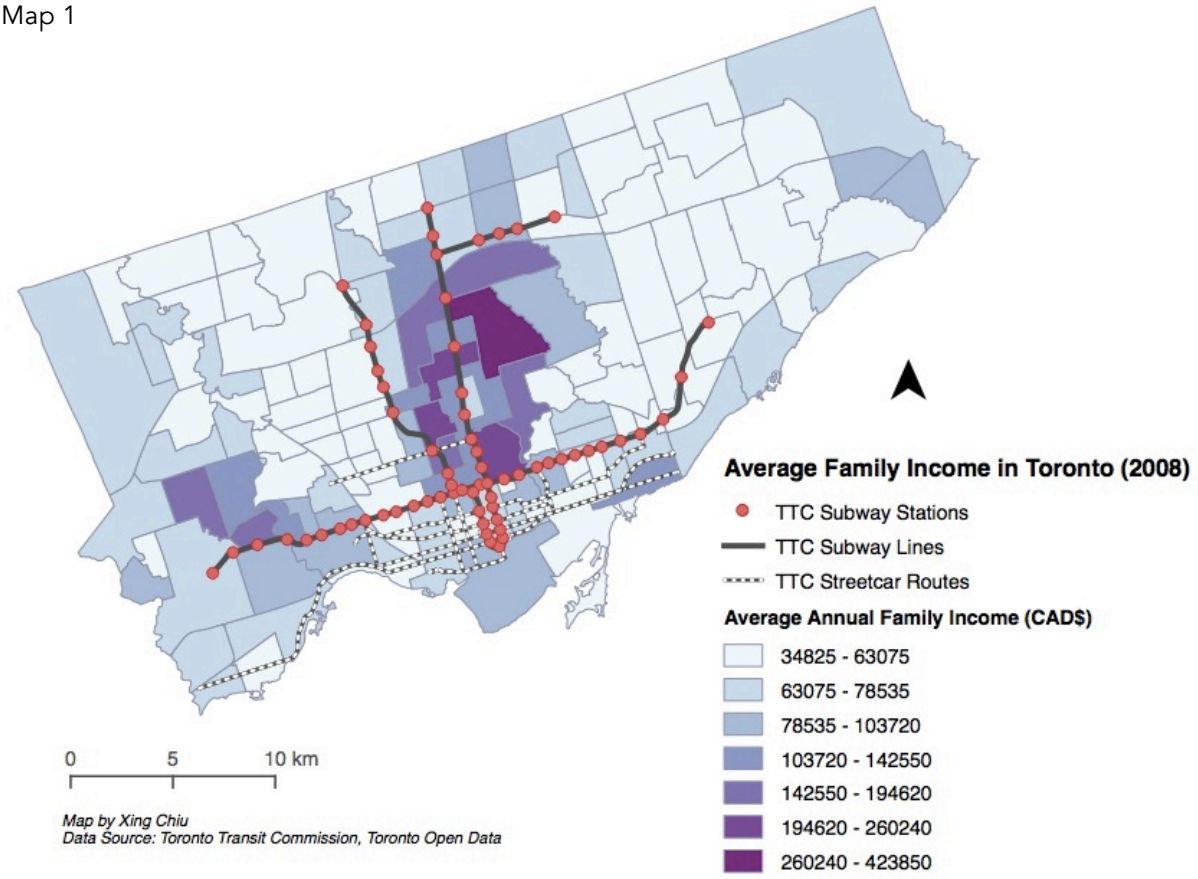
The city of Toronto is situated on the northwestern shore of Lake Ontario in the densely populated region of southern Canada known as the Golden Horseshoe. It has a total land area of 630 squared kilometres (Careless, 2014) and as of 2012 a population of nearly 2.8 million, making it the largest city in Canada in terms of size and population. The city's cosmopolitan and multicultural nature is a reflection of its role as an important destination for immigrants to Canada, with non-Canadian-born residents making up nearly 50% of the overall population (Careless, 2014). Toronto is also Canada's commercial capital, with the Toronto Stock Exchange and the nation's five largest banks located in the city. In terms of public transportation, the city is serviced by the Toronto Transit Commission (TTC), a publicly funded agency which operates buses, streetcars, and rapid transit (subway) services. Presently, the TTC is comprised of four rapid transit lines with 69 stations, over 149 bus routes, and 11 streetcar lines (ttc.ca).

Mapping Social Issues in Toronto

In 2007, the University of Toronto Cities Centre published a report entitled *Three Cities Within Toronto,* which illuminated the changing economic trends in the City of Toronto over three decades. The study found notable economic polarization and segregation within the city, and identified three geographically distinct areas within Toronto – the first where incomes increased 20%, the second where incomes have increased or decreased less than 20%, and the third where incomes decreased 20% or more since 1970 (Hulchanski, 2010). Building on the findings of the Three Cities report, the *Martin Prosperity Institute* conducted further research to investigate how the availability of public transportation affected the changes and spatial distribution of income. The Institute developed *transit scores* calculated for all census areas of the city using data about the number and proximity of stops and frequency of public transportation services. Not surprisingly, transit scores were highest in the downtown core, which is serviced by streetcars, buses, and subways, and lowest towards the outer edges of the city serviced only by busses (Martin Prosperity Institute). Their research revealed that areas underserved by the transit system, which they refer to as "transit deserts", greatly overlapped with the areas where incomes decreased by 20% in Hulchanski's Three Cities study (ibid). This highlighted the likely significant role public transit plays in a neighbourhood's socio-economic wellbeing.

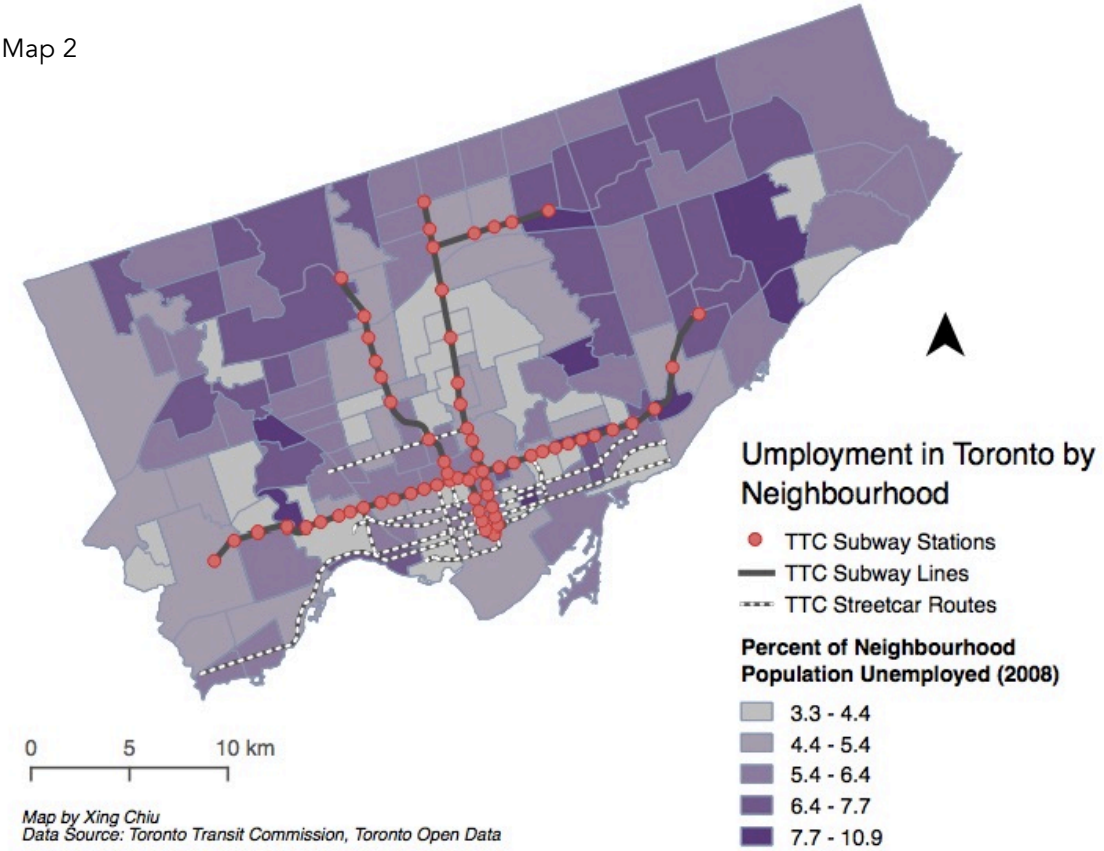
Using 2008 census data obtained from *Toronto Open Data,* I found that there is still large geographically-tied income disparity in Toronto. *Map 1* shows the average annual family income in each of Toronto's 140 officially recognized neighbourhoods with the Toronto public transit route overlaid. As can be seen, average family incomes range from $34,825 (CAD) to $423,850 (CAD) per year, with higher income families living in neighbourhoods well connected by rapid transit and lower income families living mostly in the city's inner suburbs, which are solely serviced by busses. This supports the findings of the Martin Prosperity Institute and illustrates the important impact that available public transit has on family income.

Map 1

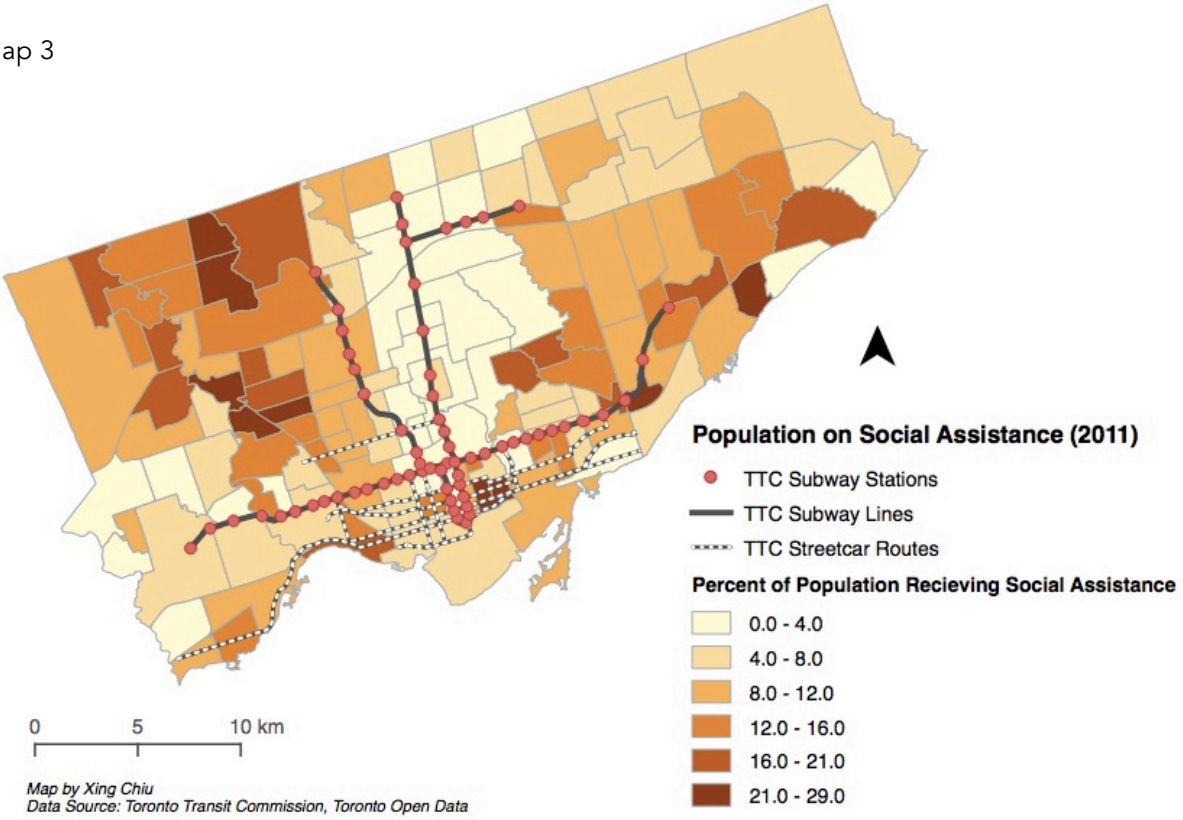


It is important to understand that this is not a cause and effect relationship, i.e., the lack of rapid transit is not a cause of lower-income but rather perpetuates a vicious cycle: lower-income individuals and families tend to live in the inner-suburbs because housing is more affordable than in the city centre. The reason that housing is relatively cheaper is in a large part due to the lack of available public transit connections, employment opportunities, and services. Toronto's inner suburbs were planned for middle-income families in single dwelling homes that had access to a car. As such, these areas were designed with limited retail developments, employment opportunities, and social services within walking distance. Many low-income families who now reside there do not own vehicles, and with the inaccessible public transit (both in terms of availability and affordability), often have a difficult time accessing employment and other necessary services (health, education, etc.) to improve their wellbeing. Unable to access work and social services, many individuals and families are isolated in their communities, face high unemployment rates, rely on social assistance, and cannot break out of their cycle of poverty.

Map 2

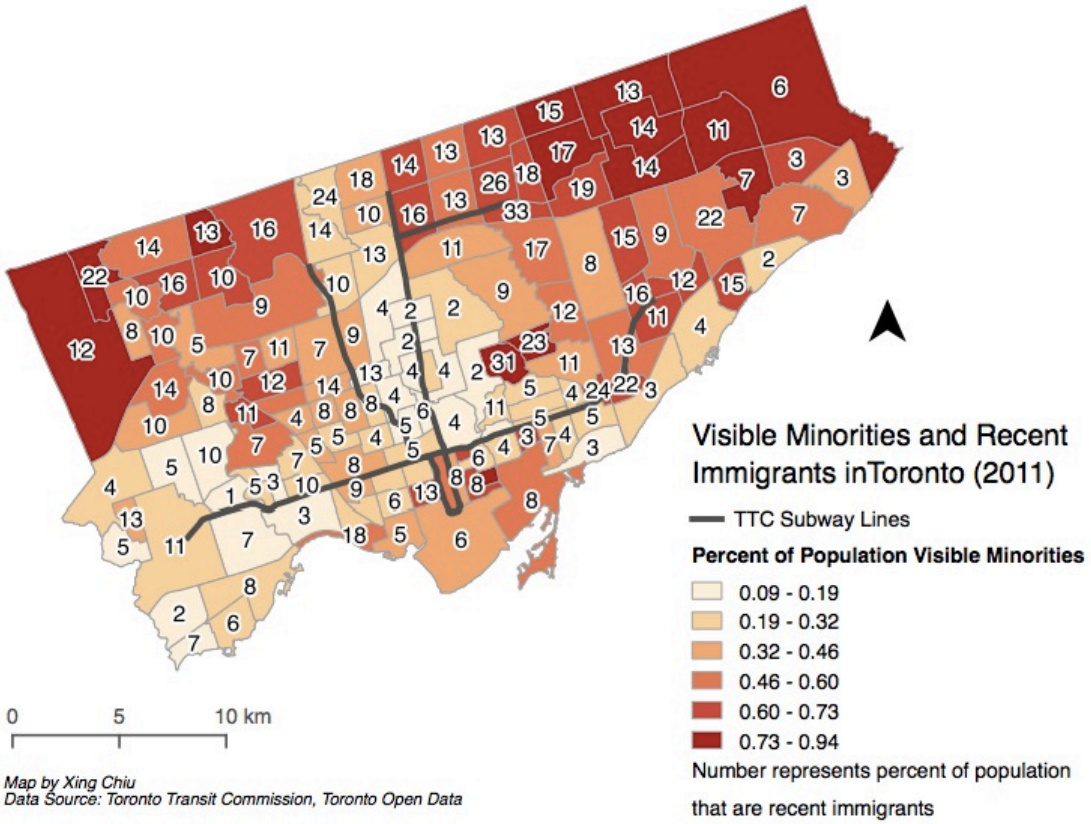

Umployment in Toronto by Neighbourhood
TTC Subway Stations
TTC Subway Lines
TTC Streetcar Routes
Percent of Neighbourhood Population Unemployed (2008)
3.3 - 4.4
4.4 - 5.4
5.4 - 6.4
6.4 - 7.7
7.7 - 10.9
0
5
10 km
Map by Xing Chiu
Data Source: Toronto Transit Commission, Toronto Open Data

Map 3


Population on Social Assistance (2011)
TTC Subway Stations
TTC Subway Lines
TTC Streetcar Routes
Percent of Population Recieving Social Assistance
0.0 - 4.0
4.0 - 8.0
8.0 - 12.0
12.0 - 16.0
16.0 - 21.0
21.0 - 29.0
0
5
10 km
Map by Xing Chiu
Data Source: Toronto Transit Commission, Toronto Open Data

However, socio-spatial exclusion is not just a matter of lower income levels, unemployment, and reliance on social assistance. It is also closely related to ethnic segregation.

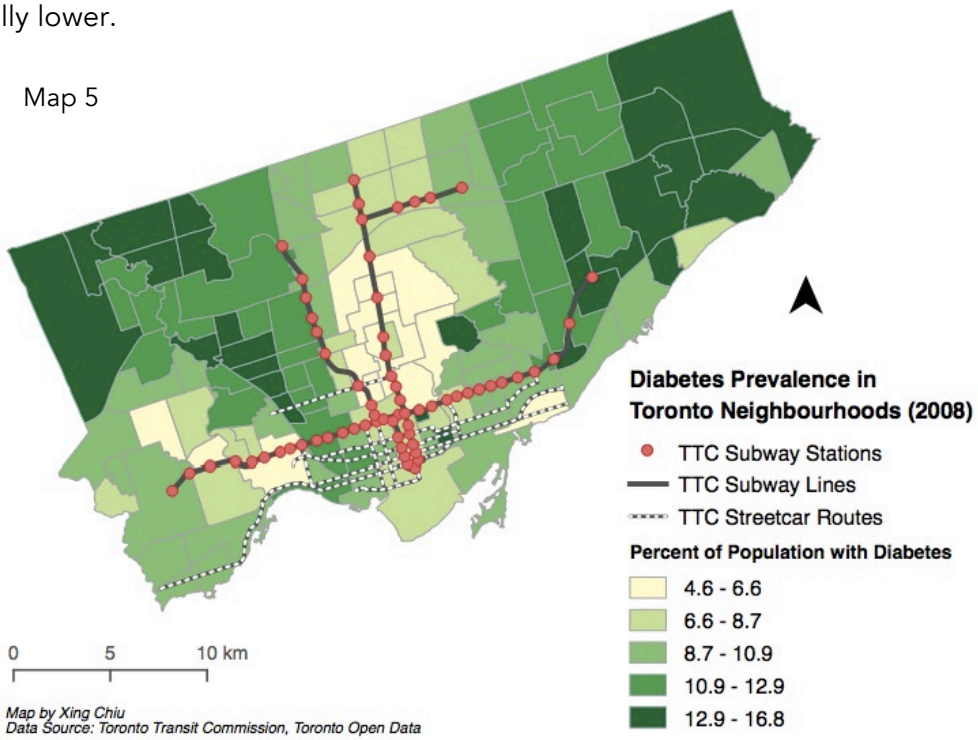
Map 4



While some immigrants choose to settle in areas with others of a similar cultural background or immigrants in general, a disadvantage of ethnic enclaves is that "they can be isolating, inhibiting immigrants' acculturation to the ways of Canadian job market and social mores" (Qadeer, 2003). The socio-spatial marginalization of recent migrants and visible minorities makes it more difficult for them to participate in society and asserting their rights. Moreover, segregation is often associated with and reinforces the cycle of poverty: A *United Way of Toronto* report, 'Poverty by Postal Code' for instance, suggested that the growth in visible minority families was highly correlated with the growth in family poverty within the City of Toronto between 1981 and 2001 (Walks, 2006).

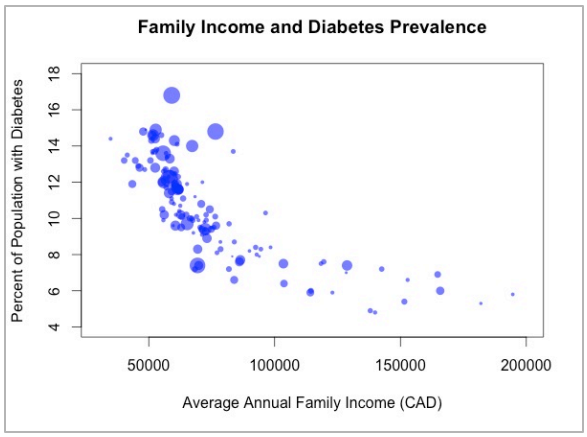
Socio-spatial isolation is also related to health issues. The map below shows the prevalence of diabetes within different neighbourhoods across the city. It is clear that diabetes is much more prevalent in areas beyond the reach of the rapid transit line and where incomes are generally lower.

Map 5



Indeed, there seems to be a correlation between a neighbourhood's average income and its diabetes prevalence (see *Graph 1*): as incomes increase, the percent of the population with diabetes decreases. Each circle represents one neighbourhood, sized relative to its population.

Graph 1: Income and Diabetes Prevalence in Toronto



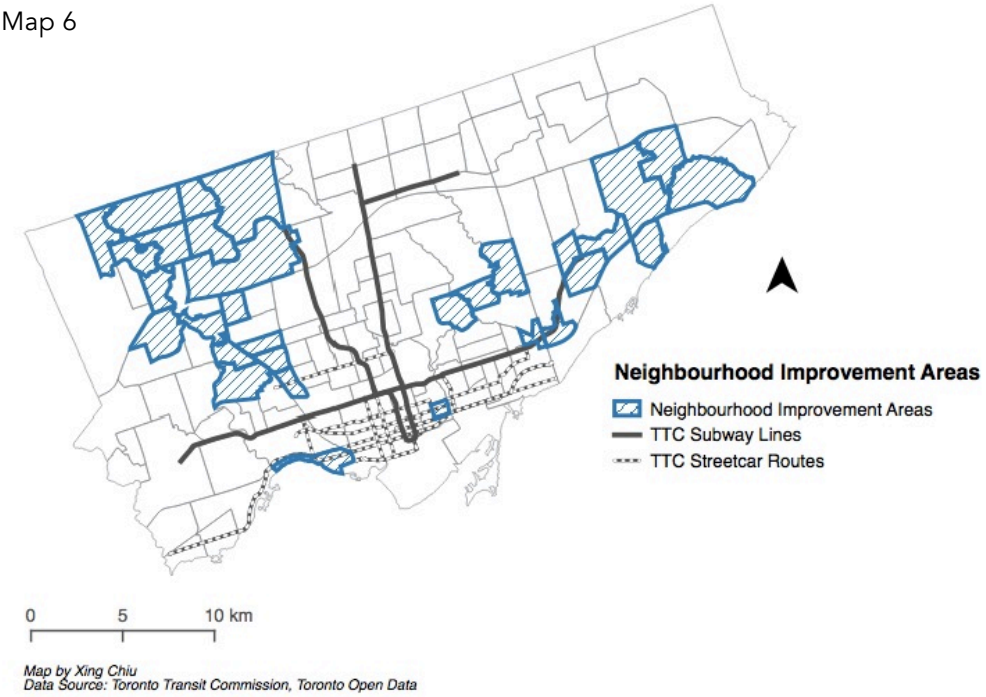
Made by Xing Chiu, Data source: Toronto Open Data (2011 Census)

This is not to suggest that increasing a neighbourhoods wealth reduces diabetes prevalence; but the trend does imply that socio-spatial isolation is likely related to and perhaps a contributing factor that perpetuates diabetes and other health issues. It is significantly more difficult for low-income families in socio-spatially isolated neighbourhoods to obtain the proper health care they need, and maintaining good health (including successful diabetes management) requires regular access to health care services. Improving the availability of public transit in parts of the city with high diabetes rates could increase access to services necessary for the diagnosis and control of diabetes and its related conditions. It is also crucial that the public transit is economically accessible particularly for the lower-income individuals in neighbourhoods where diabetes is most prevalent. The City's Medical Officer for Health suggested providing discount tickets and incorporation of transportation costs in social assistance rates in a report to City Hall as potential actions (City of Toronto 2013).

Steps Forward

In March 2014, the City identified 31 of Toronto's neighbourhoods as "at risk" areas for falling below a benchmark *equity score* (comprised of 15 socio-economic factors including education level, unemployment rate, crime, and health issues). As can be seen on the Map of Neighbourhood Improvement Areas on the following page, the majority of the priority neighbourhoods are in the inner suburbs of the city where poverty, visible minorities, recent immigrants, and diabetes are most prevalent. These priority neighbourhoods will receive special attention by the municipality in order to strengthen them as part of the City's Strong Neighbourhoods Strategy 2020. The City has set out plans to improve and revitalize these areas through "by partnering with residents, businesses and agencies to invest in people, services, programs and facilities in specific neighbourhoods." (City of Toronto) While these are very important steps, it is critical that the City also significantly augments accessibility to public transportation, which is likely one of the main contributing factors to the continued social and economic exclusion of these areas.

Map 6



Overlaying public transit routes across the city makes evident how the lack of accessibility to public transit is exacerbating socio-spatial exclusion in Toronto. Hence, in order to break vicious cycles of segregation and poverty and to enhance the sustainability of the city in general, improvements to make the public transit system more accessible are necessary. First, this entails greatly augmenting availability of public transit to pockets of the city, namely the inner-suburbs, which are currently underserved by the TTC. Ideally, the rapid transit (subway line) would be extended into these neighbourhoods. If this is financially impossible for the municipality (which is the likely case at least in the near future), providing much more frequent bus services would be a necessary start. Second, public transit should be made more affordable to lower-income individuals: Current adult monthly pass rates range from $129.75 (under discount plan for yearly subscription) to $133.75 (regular) (ttc.ca). This accounts for 7.1% of monthly income for those earning minimum wage, making it the second most expensive public transit service in the country (City of Toronto, 2013). The government should provide discounts to low-income individuals in order to facilitate their mobility. Socio-spatial exclusion prevents a city from developing in a healthy and sustainable manner. In Toronto, improving the accessibility of the public transit system can help break vicious cycles of poverty linked to socio-spatial exclusion and thus should be a priority for the City.

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